MICROSOFT AZURE · PRODUCTION ARCHITECTURE

prodDisruption — Flight Disruption Intelligence Platform

A multi-tenant, AI-driven SaaS platform on Azure that automates passenger recovery and personalised notifications during flight disruptions, built on Azure Container Apps, Azure AI Foundry, Azure API Management and Azure Cosmos DB.

OWNER	CLOUD	COMPUTE
AIONOS Platform Engineering	Microsoft Azure	Azure Container Apps (×3) + Static Web Apps

AI RUNTIME	EDGE	DOCUMENT VERSION
Azure AI Foundry	Azure API Management	1.0

1. Executive Summary

1.1 What is prodDisruption

prodDisruption is a multi-tenant, AI-driven SaaS platform that automates how an airline handles passenger experience the moment a flight is **cancelled** or **delayed**. It sits in front of the airline's existing CRM, Customer Data Platform (CDP) and flight inventory systems, and turns every disruption event into one of two outcomes:

- **Recovery** — when a flight is cancelled, an Azure AI Foundry agent selects the best alternate flight and the best available seat for that passenger, in under 30 seconds, using rules derived from the passenger's CDP persona (Student, Doctor, Defence, High Spender, General).
- **Messaging** — when a flight is delayed, an Azure AI Foundry agent selects the right message group and produces three personalised, channel-aware notifications (SMS, WhatsApp, Email) from the airline's approved template set, again driven by the passenger's persona.

1.2 Why this is needed

Flight disruptions are one of the largest unmanaged cost and CX problems in commercial aviation:

- Globally, irregular operations (IROPS) cost the aviation industry an estimated **USD 30+ billion every year** in operational recovery, compensation and lost loyalty (industry estimates).
- A single cancellation on a narrow-body aircraft produces roughly **180 individual rebooking conversations** that today queue on contact-centre agents, with handle times in tens of minutes per passenger.
- Generic disruption SMS treats every passenger identically — a Defence traveller, a Doctor, a Student and a top-tier loyalist all receive the same template. Personalisation at scale is operationally impossible by hand.
- Airlines competing on customer experience need to **scale empathy**: every disrupted passenger needs an immediate, individually-relevant response — not a position in a phone queue.

1.3 Business outcomes

Lower IROPS cost

↓ Contact-centre handle time and operational cost during disruption events

Higher NPS

Sub-minute, personalised responses replace 20-minute call queues

Loyalty retention

Auditable playbook

Persona-aware recovery protects high-value segments (High Spenders, Defence, Doctors)

Standardised, logged disruption handling — every decision is traceable

1.4 How the platform works

Each disruption request follows the same four-step flow: (1) the platform ingests the disruption event from the airline's operational systems, (2) resolves the passenger's persona via the airline's CDP, (3) invokes the tenant-scoped **Azure AI Foundry** agent for either Recovery or Messaging, and (4) returns a deterministic, structured JSON response that the airline's downstream channels (web, mobile, contact centre, OTAs) consume directly.

1.5 Engineering posture

The platform is deployed entirely on managed Microsoft Azure services: **three Azure Container Apps** (`api-prod` , `api-admin` , `mcp-server`) behind **Azure API Management**, with an **Azure Static Web Apps** front-end for tenant administration. **Azure Cache for Redis** provides idempotency and hot-path caching; **Azure Cosmos DB for MongoDB** stores tenant configuration, AI agent registry, prompts and templates; **Azure Key Vault** holds all secrets with **Managed Identity** end-to-end; **Microsoft Entra ID** issues identities for admins and inter-tenant federation. Tenant configuration and the AI agent registry are hot-swappable — agent IDs, prompts and templates change at runtime without an application restart.

Multi-tenant

Logical multi-tenant, partitioned by `tenant_id` at every layer

Deterministic AI

temperature = 0.1, top_p = 0.1, structured JSON output with hallucination guard

Zero-trust

Managed Identity, Key Vault references, private endpoints, no shared secrets

SaaS economics

Per-tenant metering, per-app autoscaling, per-tenant cost attribution

2. Business Context

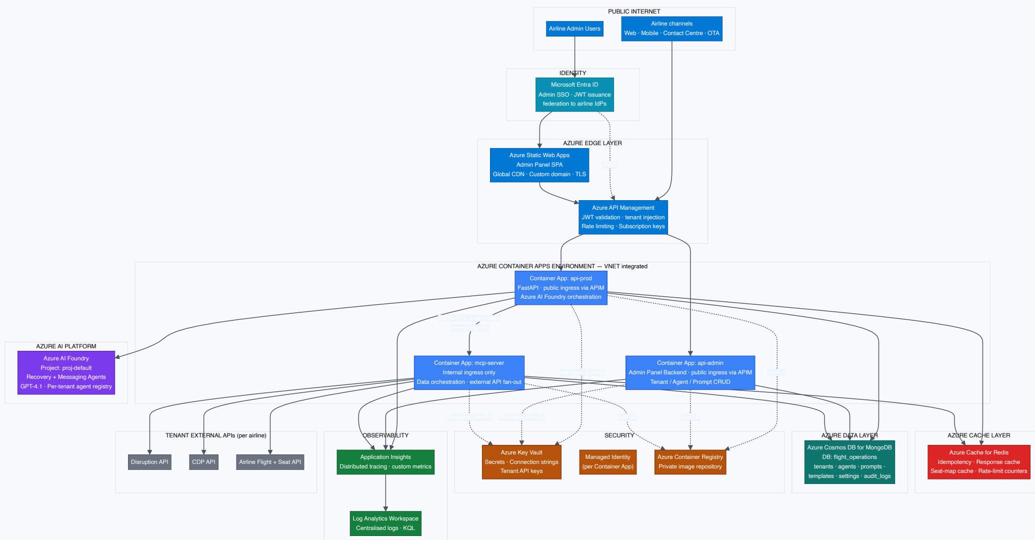
STAKEHOLDER	OUTCOME
Airline passengers	Sub-minute personalised recovery (alternate flight + seat) or proactive delay notifications
Airline operations	Automation of disruption recovery; reduction in contact-centre load
Airline marketing / CRM	Persona-aware messaging across Student, Doctor, Defence, High Spender, General segments
Airline IT	Drop-in SaaS — no on-premise AI infrastructure; standards-based REST API
AIONOS	Multi-tenant SaaS economics on Azure with per-tenant isolation and metering

Supported disruption flows

- **Recovery flow** — `event_type = flight_cancelled`. AI selects the best alternate flight + best seat based on CDP-derived priority rules.
- **Messaging flow** — `event_type = flight_delayed`. AI selects the best message group and personalises SMS / WhatsApp / Email content.

3. Logical Architecture

The platform is composed of three Azure Container Apps fronted by Azure API Management. The Admin Panel is delivered as a Static Web App. All data and identity services are accessed over private endpoints from inside the Container Apps VNET.



4. Component Responsibilities

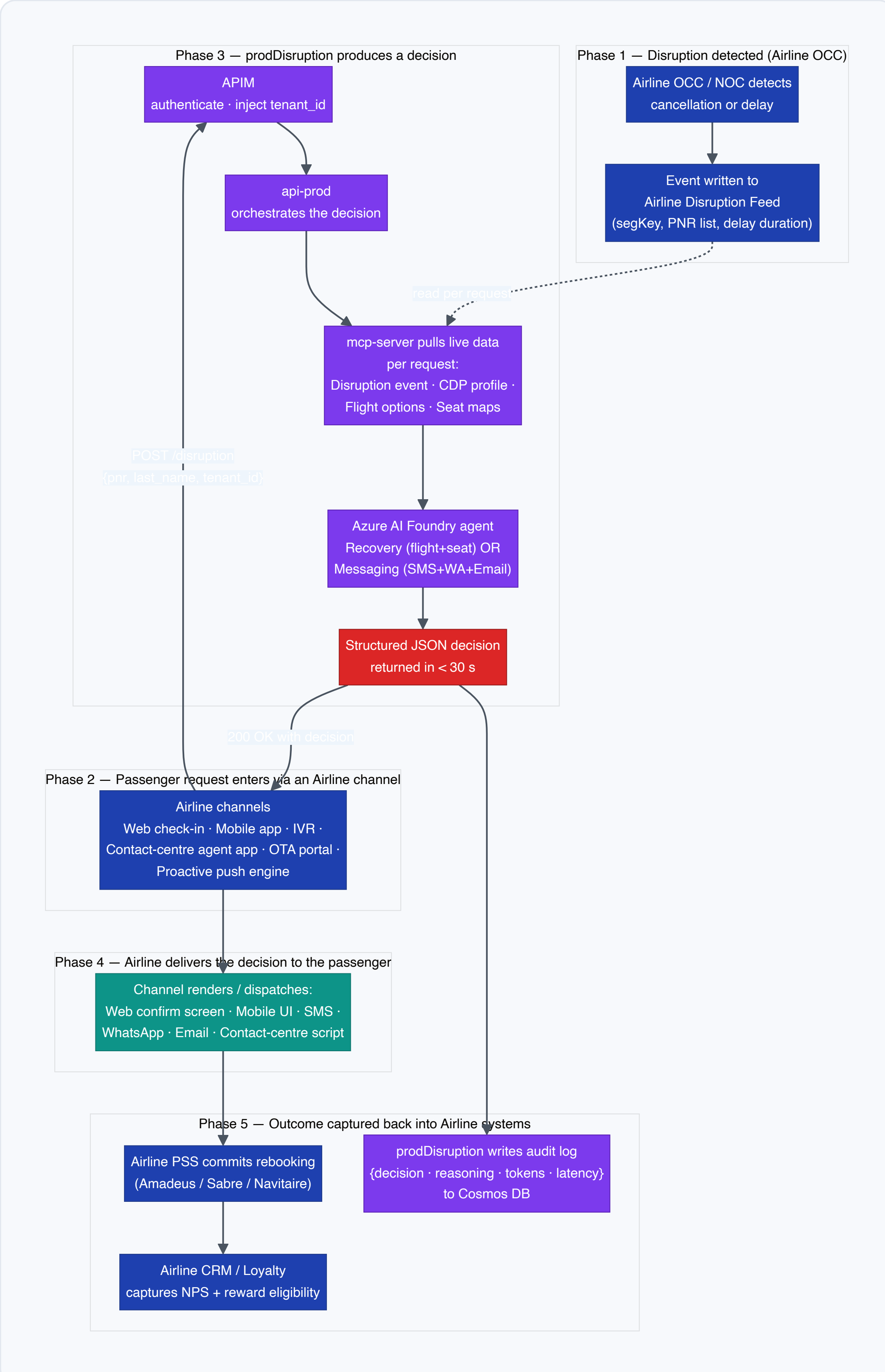
COMPONENT	AZURE SERVICE	RESPONSIBILITY	INGRESS
Admin SPA	Azure Static Web Apps	Airline admin UI for tenant config, agent registry, prompt and template management	Public (HTTPS, custom domain)
APIM	API Management Standard v2	Single ingress, JWT auth, tenant context injection (`X-Tenant-Id`), rate limiting, subscription keys, schema validation, request logging	Public (HTTPS)
`api-prod`	Azure Container Apps	Public REST API. Receives disruption requests, orchestrates the MCP call, executes the Azure AI Foundry agent and returns the structured response	External — via APIM only
`api-admin`	Azure Container Apps	Backend for the Admin SPA. CRUD on tenants, agents, prompts, templates. Writes audit logs to Cosmos DB	External — via APIM only
`mcp-server`	Azure Container Apps	Data orchestration plane. Parallel async fan-out to Disruption API, CDP API and Airline API. Resolves tenant base URLs from Cosmos DB	Internal — VNET only
AI Foundry	Azure AI Foundry (GPT-4.1)	Hosts the Recovery Agent and Messaging Agent. Per-tenant agent IDs	Managed Identity authenticated

COMPONENT	AZURE SERVICE	RESPONSIBILITY	INGRESS
		resolved at runtime from Cosmos DB	
Redis Cache	Azure Cache for Redis	Hot-path caching: idempotency keys (de-dup repeated PNR requests), response cache for completed jobs, seat-map cache (30 min TTL), tenant-config and agent-config short-TTL cache, rate-limit counters	Private endpoint (Premium tier) / firewall + MI auth (Standard tier)
Cosmos DB	Cosmos DB for MongoDB	System of record for tenant config, agents, prompts, templates, settings, audit logs	Private endpoint
Entra ID	Microsoft Entra ID (P1)	Identity provider for airline admin users (SSO into the Admin SPA), JWT issuer for APIM validation, federation point for airline IdPs (B2B), service principals for CI/CD	Public OIDC / OAuth2 endpoints
Key Vault	Azure Key Vault	Centralised secret store. All Container Apps read via Key Vault references — no plaintext secrets in app config	Private endpoint
ACR	Azure Container Registry (Premium)	Signed, scanned container images for all three apps. Image pull via Managed Identity	Private endpoint
Application Insights / LAW	Azure Monitor	Distributed tracing, custom metrics, log aggregation, alerting	n/a

5. Runtime Flows

5.1 End-to-End Operational Flow — how an airline uses prodDisruption

The diagram below shows the full lifecycle of a single disruption — from detection in the airline's Operations Control Centre (OCC) through to passenger delivery and outcome capture in the airline's PSS / CRM. prodDisruption sits between the airline's **operational data systems** (disruption feed, CDP, flight inventory) and the airline's **passenger channels** (web, mobile, contact centre, OTA, push gateway). The platform *produces decisions*; the airline's existing systems remain the system of record for rebooking, notification dispatch and loyalty capture.



5.2 Data inputs and outputs per request

Each disruption request consumes a small, strictly-defined input set and produces a strictly-shaped JSON output. The platform itself is stateless on the request path — no passenger PII is persisted; only decision metadata (no personal data) is written to the audit log.

DIRECTION	FIELD / PAYLOAD	SOURCE	PURPOSE
Inbound — request	`pnr`	Calling channel (Web / Mobile / IVR / OTA)	Lookup key for disruption event & CDP profile
	`last_name`	Calling channel	Identity verification against the PNR
	`tenant_id`	APIM (extracted from JWT claim)	Routes to correct airline tenant config + agent
Pulled — airline operational data	Disruption event	Airline Disruption API	Event type · segKey · delay / cancel context
	CDP profile	Airline CDP	Persona signals — Student · Doctor · Defence · High Spender · General
	Flight options + seat maps	Airline Flight Inventory API	Available alternates + assignable seats (recovery flow only)
Pulled — platform config	Active agent ID + admin prompts	Cosmos DB `agents` + `prompts` (per tenant + flow)	Hot-swappable AI configuration
	Message templates	Cosmos DB `templates`	Approved base content for messaging flow
Outbound — recovery	`selected_flight` + `flight_reasoning`	Azure AI Foundry —	Flight number · time · class ·

DIRECTION	FIELD / PAYLOAD	SOURCE	PURPOSE
flow		Recovery Agent	segKey · why this flight
	`selected_seat` + `seat_reasoning`	Azure AI Foundry — Recovery Agent	Seat number · travel class · seat type · why this seat
Outbound — messaging flow	`selected_group_id` + `reason`	Azure AI Foundry — Messaging Agent	Persona-matched template group identifier
	`messages[SMS, WhatsApp, Email]`	Azure AI Foundry — Messaging Agent	Personalised, channel-aware copy ready for dispatch
Side-effect	Audit log entry	Platform → Cosmos DB `audit_logs`	`{tenant_id, pnr_hash, decision, agent_id, tokens_in/out, latency_ms, ts}` — no PII

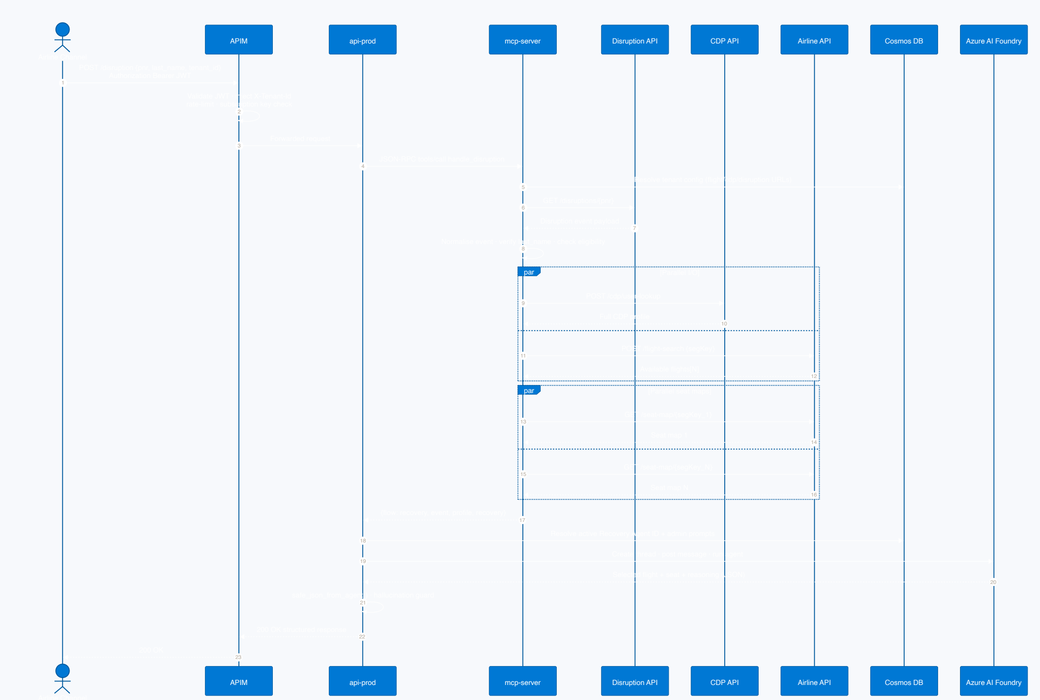
5.3 Decision contract — who owns what

prodDisruption is a **decision service**, not an execution service. It returns a JSON decision; existing airline systems remain responsible for the rebooking transaction, notification dispatch and loyalty capture. This separation keeps the integration non-disruptive to the airline's PSS and audit posture.

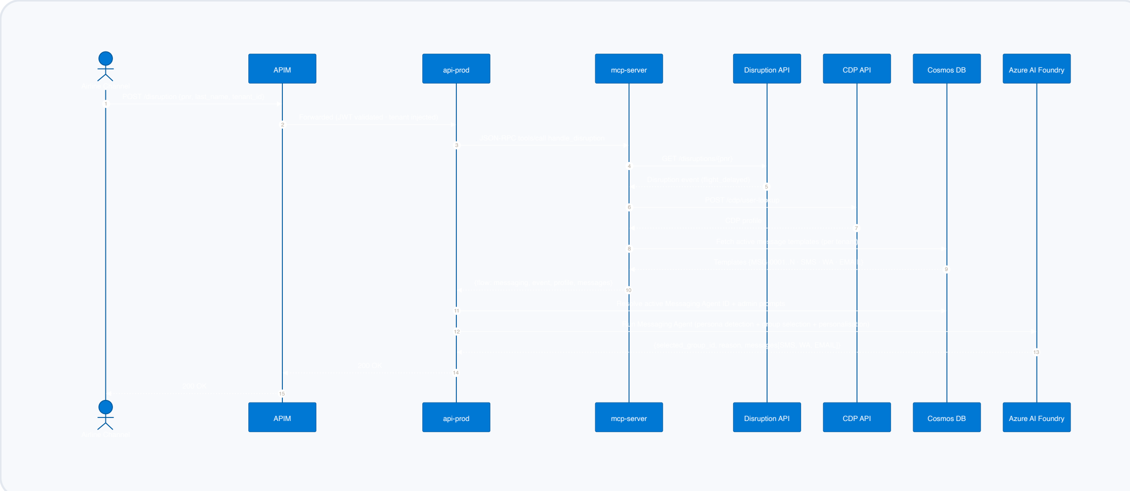
STEP	OWNER	SYSTEM OF RECORD
Disruption detection	Airline OCC / NOC	Airline FOS / OCC tooling
Disruption event publishing	Airline IT	Airline Disruption API

STEP	OWNER	SYSTEM OF RECORD
Passenger persona	Airline marketing / data	Airline CDP
Flight options + seat maps	Airline IT	GDS / NDC / Airline inventory
Decision generation (AI)	**prodDisruption**	**Azure AI Foundry agents**
Delivery to passenger	Airline channel	Channel-owned (web / mobile / SMS gateway / agent app)
Rebooking commit	Airline reservations	Airline PSS (Amadeus / Sabre / Navitaire)
Loyalty & NPS capture	Airline CRM	Airline loyalty platform
Decision audit trail	prodDisruption	Cosmos DB `audit_logs`

5.4 Recovery flow — technical sequence (cancelled flight)



5.5 Messaging flow — technical sequence (delayed flight)

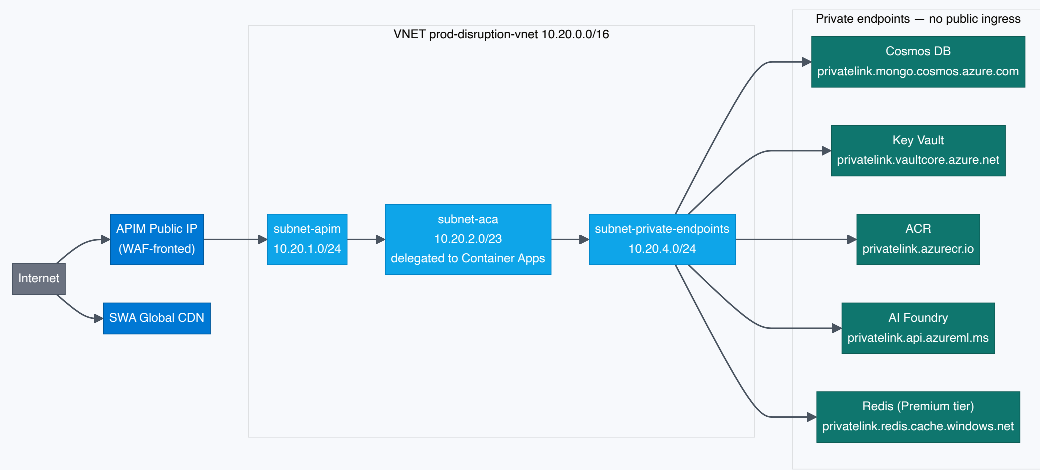


6. Azure Service Inventory

SERVICE	SKU / TIER	RATIONALE
Azure API Management	Standard v2 (Stv2)	Production multi-tenant ingress, VNET integration, subscription keys, JWT validation
Azure Container Apps	Dedicated workload profile (D4)	Per-app autoscaling, VNET integration, predictable cost
Azure Static Web Apps	Standard	Custom domain, SSO with Entra ID, staging environments
Azure Cosmos DB for MongoDB	RU autoscale (1K–10K)	Predictable cost or true elastic; session consistency; per-tenant partition key
Azure Cache for Redis	Standard C1 (1 GB) — Premium P1 recommended	Idempotency, response cache, seat-map cache; Premium adds private endpoint and VNET injection
Azure AI Foundry	Standard project — model: GPT-4.1	Hosts persistent assistants (Recovery + Messaging); per-tenant agent IDs
Microsoft Entra ID	Premium P1	Admin SSO, JWT issuance for APIM, federation to airline IdPs, conditional access
Azure Key Vault	Standard	Secrets store; HSM tier upgrade available on enterprise contracts
Azure Container Registry	Premium	Geo-replication, content trust, private endpoint, Defender scanning
Application Insights	Workspace-based	Distributed tracing, custom metrics, KQL across all apps

SERVICE	SKU / TIER	RATIONALE
Log Analytics Workspace	Pay-as-you-go	Central log sink + diagnostic settings for every resource
Microsoft Defender for Cloud	Defender for Containers, Key Vault, Cosmos DB	Posture management + runtime threat detection
Azure DDoS Protection	Network Protection (Standard)	Applied at the APIM public IP

7. Network Architecture

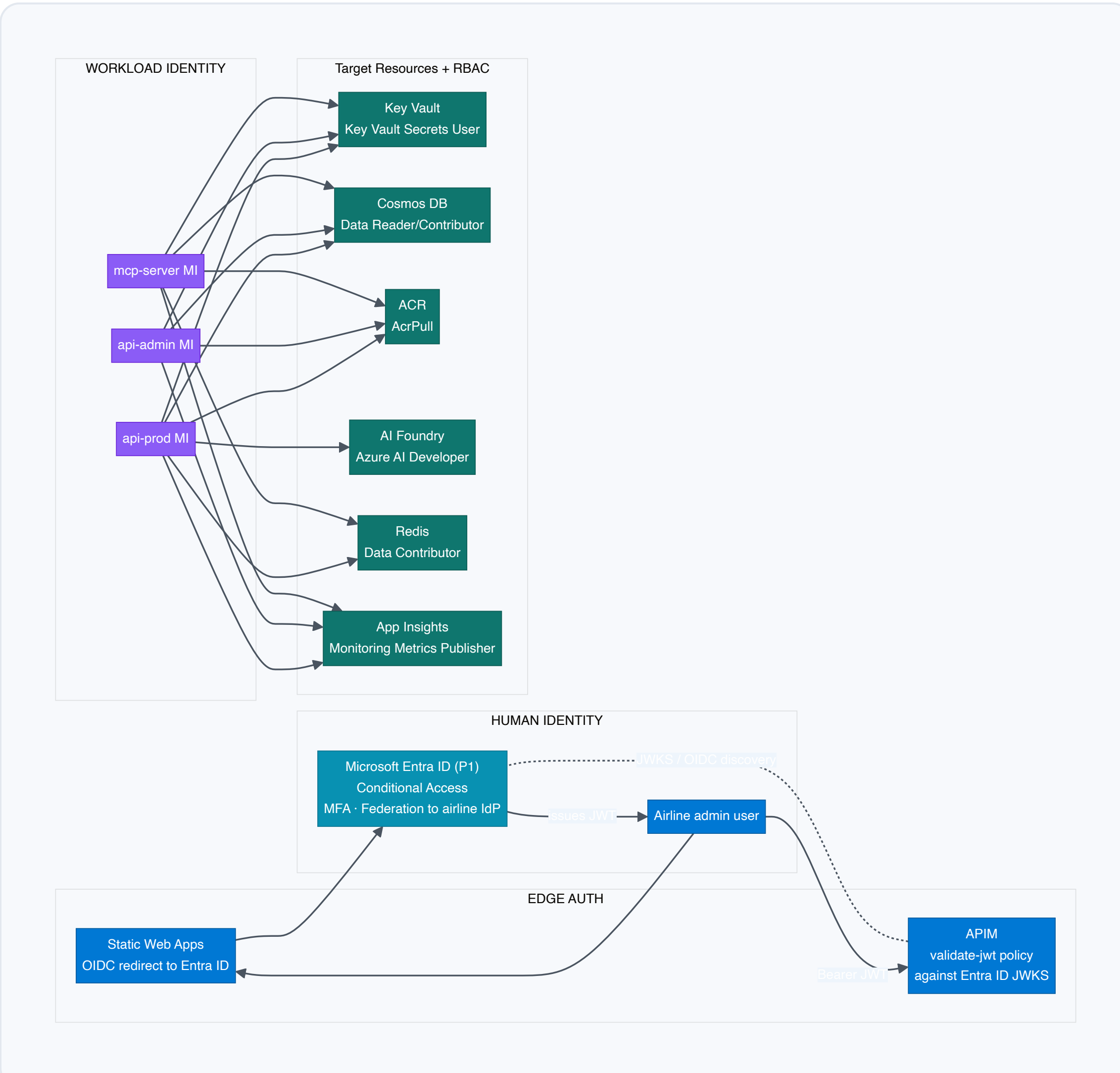


- **APIM** deployed in external mode in `subnet-apim`; egress permitted only to `subnet-aca`.
- **Container Apps Environment** is VNET-integrated. `mcp-server` uses internal-only ingress and is not addressable from the internet.
- All PaaS data services (Cosmos DB, Key Vault, ACR) are reached over **Private Endpoints**. Public network access is disabled on every account.
- Egress to tenant external APIs is routed through the Container Apps NAT — a fixed outbound public IP registered with each airline tenant for IP allow-listing.
- **NSGs** on each subnet enforce least-privilege L4 rules: APIM → ACA on 443 only; ACA → private endpoints on 443 (KV, ACR, AI Foundry), 10250 / 27017 (Cosmos), 6380 (Redis TLS).

8. Identity & Access

8.1 Identity providers and trust flow

Microsoft Entra ID is the identity authority for the platform. Two distinct identity flows are in use: **human identities** for airline admin users into the Admin Panel, and **workload identities** (system-assigned Managed Identity) for service-to-service authentication.



8.2 Zero-trust controls

PRINCIPLE	IMPLEMENTATION
No secrets in code or env	Every secret is a Key Vault reference (`secretref:` in Container Apps). Rotated via Key Vault rotation policies.

PRINCIPLE	IMPLEMENTATION
Managed Identity everywhere	All inter-service calls authenticate via system-assigned MI. No SAS tokens, no connection strings.
RBAC, least privilege	Each MI holds the minimum role required — e.g. `mcp-server` is Cosmos Data Reader, not Contributor.
Human identity	Microsoft Entra ID (P1) — Conditional Access policies enforce MFA; airline tenants may federate their own IdP via B2B collaboration.
JWT at the edge	APIM `validate-jwt` policy verifies tokens against the Entra ID JWKS endpoint. Tenant claim drives `X-Tenant-Id` header injection.
Admin Panel SSO	Static Web Apps redirects unauthenticated users to Entra ID; admin roles enforced in SWA route config and in the `api-admin` backend.
Audit logging	Every admin write creates an audit log entry in Cosmos DB. Diagnostic settings on Key Vault and Cosmos DB stream to Log Analytics.

9. Data Architecture

9.1 Cosmos DB collections (database `flight_operations`)

COLLECTION	PURPOSE	PARTITION KEY	INDEXED FIELDS
`tenants`	Per-tenant base URLs and metadata	`tenant_id`	`tenant_id`, `is_active`
`agents`	AI Foundry agent IDs per tenant per flow	`tenant_id`	`tenant_id`, `flow`, `is_active`
`prompts`	System/user prompt overrides per tenant per flow	`tenant_id`	`tenant_id`, `flow`, `is_active`
`templates`	Message templates (delay flow)	`tenant_id`	`tenant_id`, `is_active`
`settings`	Generic runtime settings	`tenant_id`	`tenant_id`
`audit_logs`	Admin write audit trail	`tenant_id`	`tenant_id`, `timestamp`

9.2 Data classification & residency

CLASS	EXAMPLES	RESIDENCY	RETENTION
Tenant configuration	Base URLs, agent IDs, prompts, templates	Primary Azure region (tenant-selected)	Lifetime of tenant
PII (transient)	PNR, last name, contact (request body only)	In-memory only, never persisted	None

CLASS	EXAMPLES	RESIDENCY	RETENTION
Audit logs	Admin actions	Same region as tenant	365 days (configurable)
Telemetry	Request traces, AI token counts	Same region as tenant	30 days hot / 90 days archive

Key guarantee: the platform does not persist passenger PII. All PII flows through `mcp-server` in memory and is discarded after the response is returned.

9.3 Consistency & backup

- **Consistency:** Session consistency (per-tenant partition reads are read-your-writes).
- **Backup:** Continuous backup with 30-day point-in-time restore.
- **Geo-replication:** Available via secondary read region with auto-failover.

9.4 Redis cache key schema

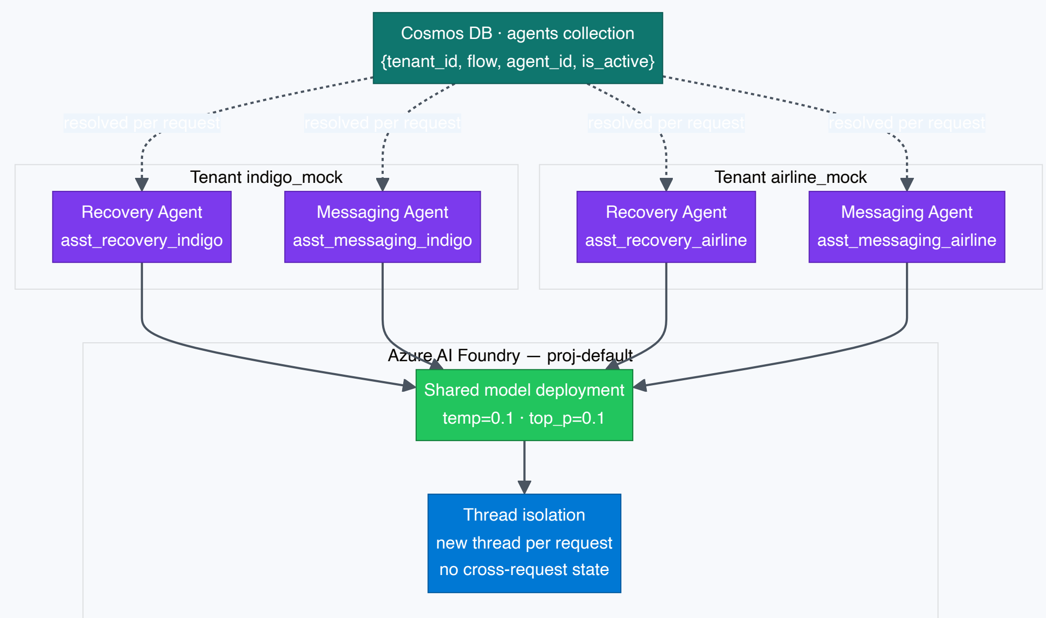
Azure Cache for Redis sits on the hot path and is keyed strictly by `tenant_id` to enforce isolation. Keys, TTLs and use-cases:

KEY PATTERN	TTL	PURPOSE
`idem:{tenant_id}:{pnr}`	300 s	Idempotency — duplicate PNR requests within the window return the cached result instead of re-running the AI agent
`result:{tenant_id}:{request_hash}`	3600 s	Response cache for completed disruption decisions
`seatmap:{tenant_id}:{segKey}`	1800 s	Parsed seat map cache — avoids re-fetching the airline seat-map API for the same flight segment
`agent_cfg:{tenant_id}`	60 s	Resolved agent IDs + active prompts from Cosmos — short-TTL to allow admin hot-swap to propagate quickly
`tenant_cfg:{tenant_id}`	300 s	Tenant base URLs for external APIs — reduces Cosmos reads on the hot path

KEY PATTERN	TTL	PURPOSE
`rl:{tenant_id}:{window}`	60 s sliding	Rate-limit counter — used by APIM and as a defensive secondary check in `api-prod`

Tier note: Standard C1 satisfies the cache workload at current volumes. Upgrading to Premium P1 enables private endpoint support, VNET injection and zone redundancy — recommended once contractual zero-trust requirements are signed with the first enterprise tenant.

10. AI Architecture



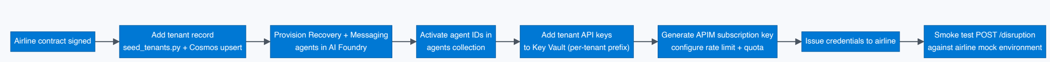
PROPERTY	VALUE
SDK	`azure-ai-projects` + `azure-ai-agents`
Authentication	`DefaultAzureCredential` — Managed Identity in production
Temperature / top_p	0.1 / 0.1 (near-deterministic)
Thread lifecycle	Created per request, never reused
Timeout	120 second polling window
Output parsing	`safe_json_from_agent()` strips markdown fences and validates JSON
Hot-swap	Activating a new agent ID in Cosmos DB applies on the next request — no app restart

PROPERTY	VALUE
Prompt layering	Base prompt → admin system prompt prepended → admin user prompt appended

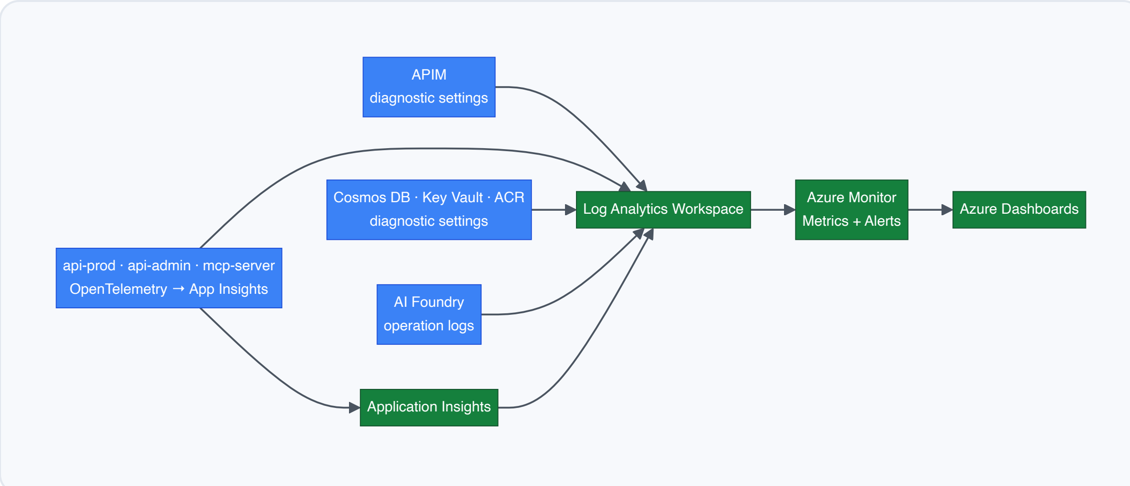
11. Multi-Tenancy Model

LAYER	ISOLATION MECHANISM	SHARED INFRASTRUCTURE?
APIM	JWT `tenant_id` claim → `X-Tenant-Id` header injection; per-tenant rate limit and quota	Shared gateway
Container Apps	Tenant context flows in headers; no in-memory tenant state	Shared compute
MCP server	`tenant_id` resolves every external API base URL	Shared compute
Cosmos DB	Partition key = `tenant_id` on every collection	Shared account, isolated partitions
Key Vault	Per-tenant secret naming: `tenants/{tenant_id}/{secret}`	Shared vault
AI Foundry	Distinct `agent_id` per tenant per flow; thread isolation enforced by SDK	Shared project, isolated agents
External APIs	Per-tenant base URLs from `tenants` collection	Fully isolated per airline

Tenant onboarding workflow



12. Observability



Custom telemetry

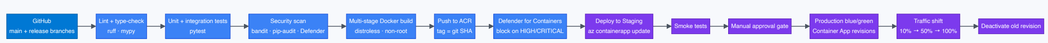
METRIC	DIMENSIONS	PURPOSE
`disruption_request_latency_ms`	`tenant_id, flow, outcome`	SLO tracking
`ai_agent_run_latency_ms`	`tenant_id, flow, agent_id`	AI performance
`ai_tokens_consumed`	`tenant_id, flow, direction`	Cost attribution
`mcp_external_api_latency_ms`	`tenant_id, api_name`	Airline-side performance
`disruption_request_total`	`tenant_id, flow, status`	Throughput & error rate

Production alert set

SEVERITY	CONDITION	ACTION
SEV 1	`api-prod` 5xx rate > 5% for 5 min	Page on-call

SEVERITY	CONDITION	ACTION
SEV 1	AI Foundry timeout rate > 10% for 5 min	Page on-call
SEV 2	p95 e2e latency > 30s for 10 min	Slack channel
SEV 2	Cosmos DB throttling (429s) > 1%	Slack channel
SEV 3	Key Vault auth failure > 0	Email
SEV 3	External tenant API error rate > 1%	Email tenant ops

13. DevOps & Release Engineering



- **Blue/green via Container App revisions** — new revision deployed with 0% traffic, smoke-tested, then progressively shifted.
- **Auto-rollback** — error-rate threshold of 1% on a new revision triggers automatic revert.
- **Hot-reload of AI config** — agent IDs, prompts and templates change in Cosmos DB without code deployment.
- **Secret rotation** — Key Vault policies rotate Cosmos DB keys (90 days) and tenant API keys. Container Apps reload Key Vault references on rotation.

14. Reliability & Disaster Recovery

The platform is deployed in a single Azure region (**South India**) with zone redundancy across all stateful services. Availability is achieved through Availability Zone distribution and Container Apps revision-level redundancy; durability is achieved through Cosmos DB continuous backup and ACR geo-replicated images.

ASPECT	CONFIGURATION
Region	South India — primary and only deployed region
Availability Zones	Container Apps Environment, Cosmos DB, APIM and Redis are deployed across the AZs available in South India
HA within region	Container Apps revisions, min replicas ≥ 2 on every app, zone-redundant SKUs
Cosmos DB	Zone-redundant; continuous backup (30-day point-in-time restore)
APIM	Zone-redundant deployment
ACR	Premium with geo-replication — images are pre-staged in a paired region for fast cold-start during DR
Key Vault	Soft delete + purge protection enabled (90 days)
Redis	Standard C1 (in-region HA via primary/replica) — Premium upgrade adds zone redundancy and geo-replication
RTO target (in-region failure)	≤ 5 minutes — AZ failover is automatic across managed services
RTO target (regional failure)	≤ 4 hours — manual restore from Cosmos PITR + ACR replica + redeploy of Container Apps via IaC
RPO target	≤ 5 minutes (Cosmos continuous backup)

Disaster recovery plan: Infrastructure is described as Bicep/Terraform; the platform can be re-provisioned in a paired Azure region from source control and Cosmos PITR in under 4 hours. A documented DR runbook covers failover steps, DNS update, tenant communication and data validation.

15. Performance & Scaling

LAYER	SCALING RULE	MIN / MAX
APIM	Fixed capacity units (Stv2)	1 → 4 units
`api-prod`	HTTP concurrency (~50 / replica)	2 → 20 replicas
`api-admin`	HTTP concurrency	1 → 5 replicas
`mcp-server`	CPU + HTTP concurrency	2 → 10 replicas
Cosmos DB (RU mode)	Autoscale 1K → 10K RU/s per collection	n/a
AI Foundry	Quota-bounded; per-region TPM / RPM monitored	Negotiated quota

Hot-path latency budget (recovery, p95)

STAGE	BUDGET
APIM	< 50 ms
MCP fan-out (parallel)	< 4 s (dominated by airline APIs)
AI Foundry recovery run	< 15 s
Total end-to-end	< 20 s p95

16. Security Controls

CONTROL FAMILY	IMPLEMENTATION
Network	VNET-integrated ACA, private endpoints for Cosmos / Key Vault / ACR; APIM is the only public ingress for the API; SWA is the only public ingress for the Admin UI
Identity	System-assigned Managed Identity per Container App; RBAC at minimum role; no SAS or connection strings
Secrets	Key Vault references only; soft delete + purge protection; rotation policies
Edge	APIM JWT validation, subscription keys, per-tenant rate limit and quota, schema validation, DDoS Network Protection on the public IP
Data in transit	TLS 1.2+ enforced; mTLS optional on internal ingress
Data at rest	AES-256 with Microsoft-managed keys; CMK upgrade path available
PII	No passenger PII persisted; in-memory only inside `mcp-server`
Threat detection	Microsoft Defender for Containers, Cosmos DB, Key Vault, CSPM
Vulnerability management	ACR + Defender for Containers; CI gate blocks HIGH / CRITICAL CVEs
Supply chain	Multi-stage builds, distroless base image, non-root runtime, signed images
Compliance posture	Aligned to ISO 27001 / SOC 2 controls inherited from Azure; per-tenant audit logs retained 365 days

17. Well-Architected Framework Alignment

Reliability

Zone-redundant SKUs · ACA min replicas ≥ 2 · Cosmos PITR · blue/green revisions · regional failover

Security

Managed Identity everywhere · Key Vault references · private endpoints · JWT at the edge · Defender for Cloud

Cost Optimisation

Autoscale on every Container App · Cosmos autoscale RU · per-tenant cost attribution tags · LAW sampling

Operational Excellence

CI/CD with security gates · blue/green · custom telemetry · production alert set · audit logging

Performance Efficiency

Async parallel fan-out in MCP · deterministic AI agents · parallel seat-map retrieval · concurrency-based autoscale